

#DC MOTOR

→ Electrical Machine that converts Electrical Energy to Mechanical Energy.

→ Achieved on the basis of Electromagnetism.

Basic Working Principle

→ When an electric current flows through a conductor placed in a magnetic field, it experiences a force.

This is known as Lorentz force.

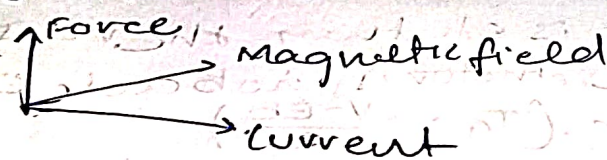
Torque Production

In a DC motor the coil (armature) is placed between 2 magnetic poles (North and South). When i passes through coil, the magnetic field interacts with current creating a force. This force generates torque, which causes the armature to rotate.

→ Force = $B \cdot I \cdot L$

B → Magnetic Flux Density
 I → Current through the Armature
 L → Length of the Conductor/coil
Unit: Newton

Fleming's Left Hand Rule



$$\vec{F} = \int d\vec{F} = I d\vec{l} \times \vec{B} = \int_0^L I d\vec{l} B \sin\theta$$

2 Arms will experience force and 2 will not.

Torque = $\tau = 2 \times (R \times F)$
 $= 2 B I L R \sin\alpha$

→ $F = B L \sin\theta I$ ($\sin\theta = 1$)
→ $F = B I L$

$\tau = 2 B I L R$ ($\sin\alpha = 1$)
max torque

Key Components

Armature :- Coil of wire that carries the current

Field Magnets :- Permanent magnets or electromagnets that produce magnetic field.

Commutator :- Device that reverse the current direction in the armature, ensuring continuous rotation.

Brushes :- Conductive material that transfer current from the power supply to the rotating commutator.

Magnetic Interaction :- The current in the coil generates a magnetic field, which interacts with a field of the permanent magnets, which gives a force.

the commutator switches the direction of current at appropriate time, ensuring torque direction remains constant for rotation.

Rectifier

BACK EMF / Counter EMF

- In a DC motor, Back EMF is the voltage that is generated in armature coil when it rotates in the magnetic field. (Opposes)
- This is an essential phenomenon that happens naturally due to Faraday's Law of Electro magnetic Induction.
- As the armature moves through the magnetic field, it cuts the magnetic lines of force. This movement induces a voltage in the coil which opposes the applied voltage.

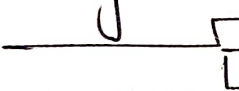
BACK EMF

→ Induced voltage opposes the input voltage that powers the motor. acc to Lenz's law, hence the word BACK EMF.

Significance

- At startup when the motor is stationary there is no back EMF hence there is no back EMF so the current is maximum. ($I_a = \frac{V - E_b}{R}$)
- As the motor speeds up, back EMF increases, the net voltage and thus the current.
- This prevents motor from drawing too much power and overheating (motor windings).

Parameters


 → B.d.A almost constant for all poly
 → more ϕ = higher magnetic field.

Energy Efficiency = Back EMF helps to regulate energy consumption.

Speed control = changes in back EMF can be used to monitor and control the motor's speed.

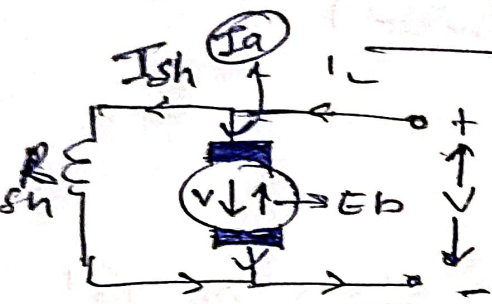
Induced EMF

$$E_b = \left[\frac{ZN\Phi}{60} \right] \left[\frac{P}{A} \right]$$

- Z = total number of armature conductors (WB)
- P = poles
- A = number of parallel paths
 - = 2 --- for wave winding
 - = P --- for lap winding (poles = P)
- N = speed of armature winding in rpm.

Armature

Current $\Rightarrow I_a = \frac{V - E_b}{R_a}$



Back EMF $\rightarrow$ Back EMF is always less than the Applied voltage

V and E_b are entry opposite

Armature Torque Equation

Mechanical power developed = $P_m = \frac{F \cdot v}{\text{Time for one revolution}} = \frac{\text{Work Done}}{\text{Time}}$

$$P_m \Rightarrow \frac{F \cdot 2\pi r}{\frac{60}{N}} = (F \cdot v) \frac{2\pi}{60} N$$

$F \cdot v = \tau \Rightarrow \left(\frac{\tau \cdot 2\pi N}{60} \right) \text{ watts}$

So, $P_m = P_e$

$V = E_b + I_a R_a$ (voltage eqn.)

$I_a = \frac{E_b}{R_a} + \frac{I_a^2 R_a}{E_b}$ $\rightarrow$ Heat Energy/Power.

$\Rightarrow \frac{\tau \cdot 2\pi N}{60} = \frac{ZN\Phi}{60} \left(\frac{P}{A} \right) I_a$

$\rightarrow$ Electrical Power (P_e)

$\Rightarrow \tau = \frac{1}{2\pi} \Phi I_a \left(\frac{PZ}{A} \right)$

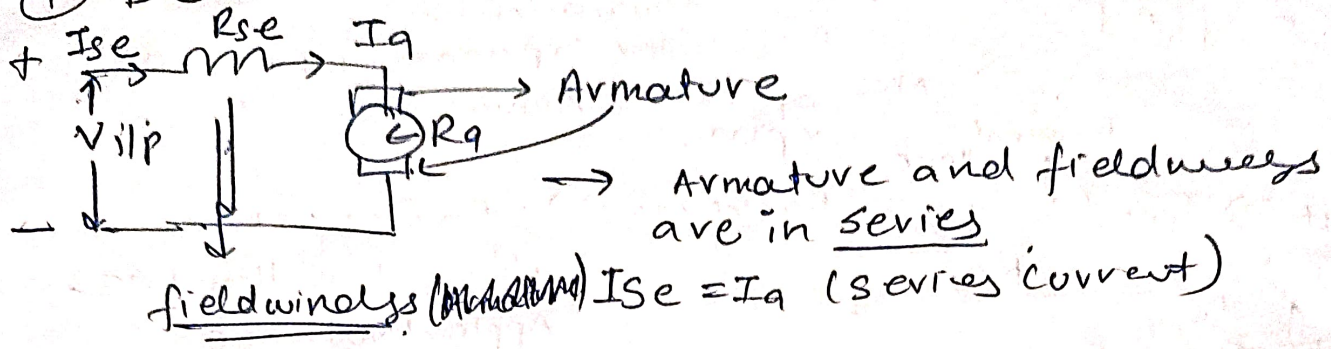
depends on type of winding.

$\Rightarrow$ Torque $\Rightarrow 0.159 \Phi I_a \left(\frac{PZ}{A} \right)$

$\rightarrow$ lap winding = No. of poles
 $\rightarrow$ wave winding = 2
 for 11 path.

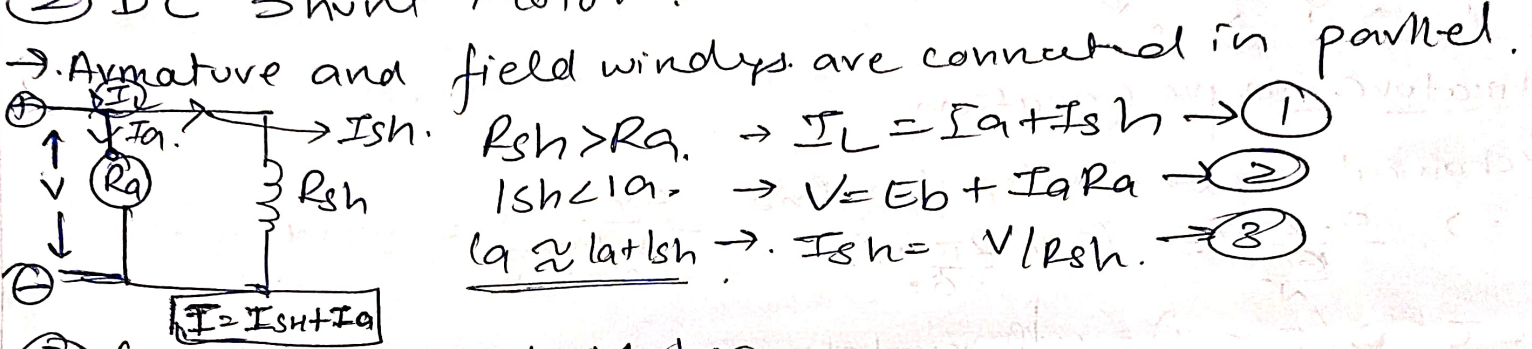
① DC Series Motor

field



$$\Rightarrow V = E_b + I_a(R_a + R_{sc}) \text{ KVL}$$

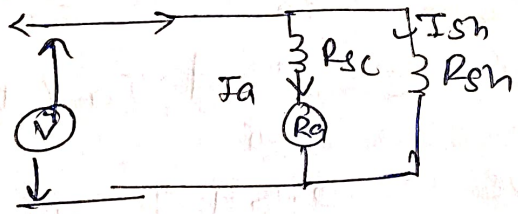
② DC Shunt Motor.



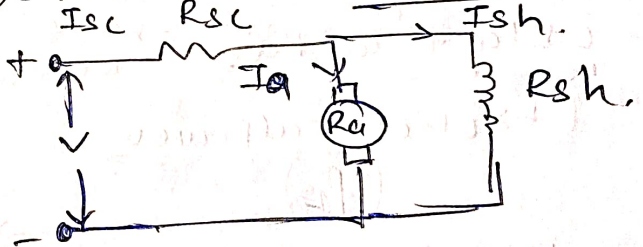
③ Compound - Wound Motor

Consist of both shunt field and series field.

(i) Long Shunt Motor



(ii) Short Shunt Motor



DC Shunt Motor

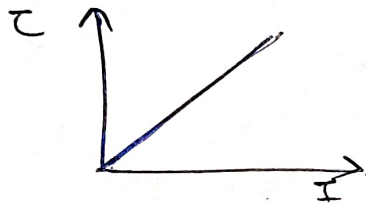
1) T_a v/s I_a Characteristic

T_a = Armature Torque ; I_a → Armature Current

$$T_a = \frac{1}{2\pi} \Phi I_a \left(\frac{PZ}{A} \right) \text{ Newton-Meter}$$

→ In DC shunt motor Φ is always constant
 $\therefore T_a \propto I_a$

$$T_a \propto I_a \Phi$$

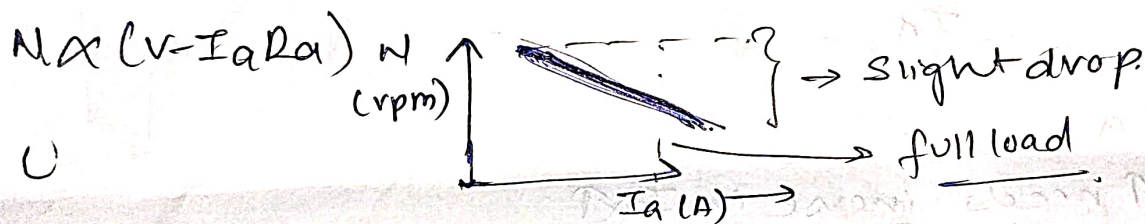


→ DC motor is a constant speed motor

② N v/s I_a

$$E_b = \frac{2N\Phi}{60} \times \frac{P}{A}$$

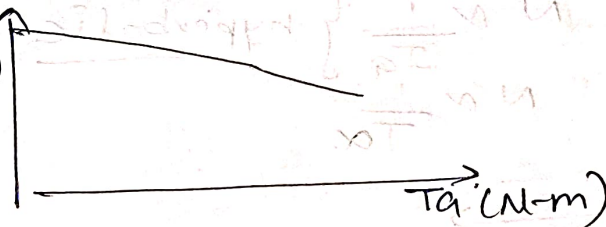
$$\frac{E_b}{\Phi} \propto N ; E_b \propto N$$



③ Speed v/s Torque N (rpm) vs T_a (N-m)

$$N \propto (V - I_a R_a) \rightarrow ①$$

$$T_a \propto I_a \rightarrow ②$$



① Constant speed motor, Elevators.

② Should not start DC shunt motor with high load

③ Φ = constant

★ DC SERIES MOTOR

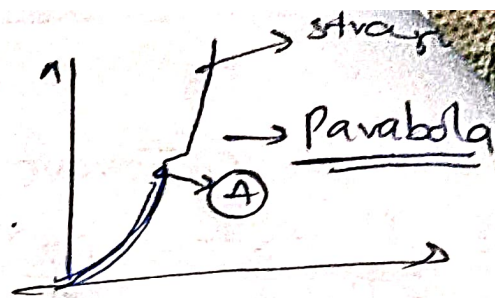
(1) Ta vs Ia

$$T_a \propto \Phi I_a$$

series motor $\Phi \propto I_a$ Armature current.

$$T_a \propto I_a I_a$$

$$T_a \propto I_a^2$$

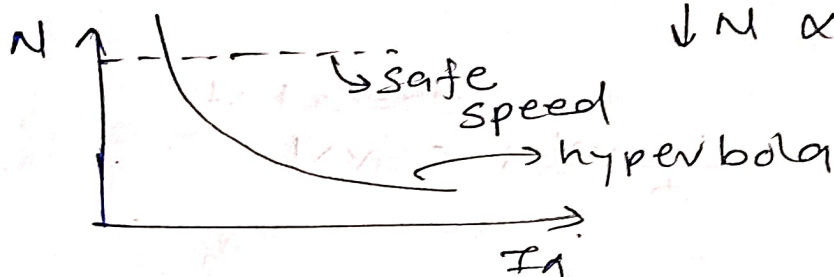


(2) N vs Ia

$$N \propto \frac{E_b}{\Phi}$$

($\Phi \propto I_a$) series motor

$$\downarrow N \propto \frac{1}{I_a} \uparrow$$



(3) N vs Ta

$T_a \uparrow$ needs increase in $I_a \uparrow$.

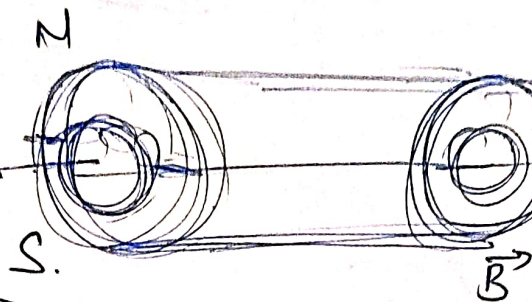
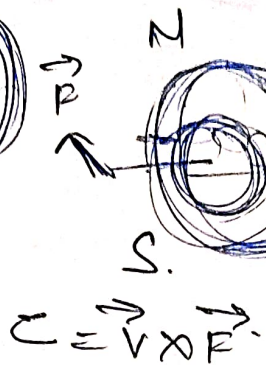
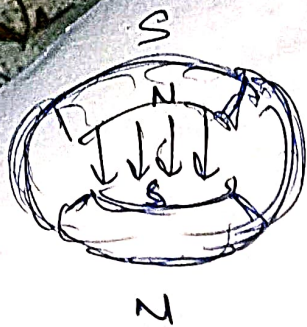
but $N \propto \frac{1}{I_a}$ } hyperbolic

$$N \propto \frac{1}{T_a}$$



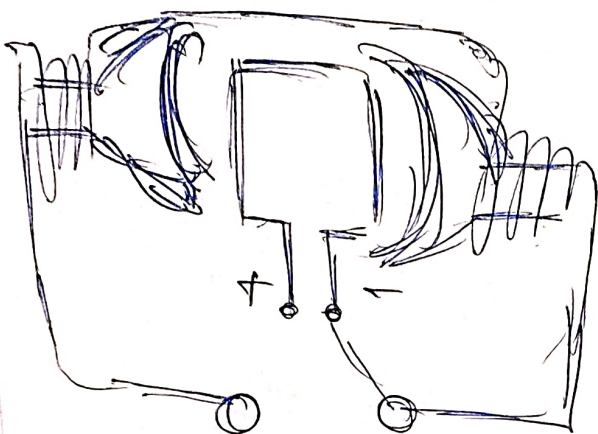
① Series Motor should start with certain load as if we won't do so, the motor will achieve dangerous speed.

② Series Motor offers high starting torque.



$$\psi = N \cdot \Phi$$

$$= N(\vec{B} \cdot \vec{A})$$



→ Diameter of the coil = D
 $R = D/2$.

→ Force experienced = $F = BIL \sin \theta$.

$$\text{Torque} = 2(\vec{R} \times \vec{F})$$

$$= (\vec{D} \times \vec{F})$$

$$|\tau| = BIL \sin \alpha$$

$$\text{Efficiency} = \eta = \frac{V_b I_a}{V \cdot I_a}$$

Instant Motor.
 $I_a \gg I_{sh}$

$$\therefore I_{sh} R \approx 0$$

$$\tau = \left(\frac{\text{Power}}{2\pi N} \right)$$

$$\tau = \frac{V_b I_a}{2\pi N}$$

$$V_b = \Phi \approx N \left(\frac{P}{A} \right)$$

$$\tau = \frac{E_b I_a \cdot 30}{N \cdot \frac{\pi}{180}}$$

$$\tau_a = \frac{E_b I_a}{N} (9.55)$$